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ABSTRACT

The dynamic behavior of nucleation and precipitation of minerals in porous media during underground fluid injection has a significant impact on many engineering applications, such as shale gas extraction and CO_2 sequestration. Traditional models usually overlook the role that homogeneous nucleation plays in this reactive flow process. Our study developed a novel numerical solver, ReactiveParticulateFlowFoam, which couples the homogeneous nucleation of minerals and the flow of reactive crystal particles at the pore scale for the first time. By simulating the reactive flow in microchannels and porous media, we found that the homogeneous nucleation behavior of minerals is governed by both the fluid flow conditions and the porous media structure. Our results indicate that an optimal Péclet number range exists that maximizes the homogeneous nucleation rate and the final amount of nuclei. In addition, the homogeneous nucleation is also affected by the porosity of the porous matrix, an increase in porosity enhances the number of nuclei, especially under advection-dominated conditions. Furthermore, we have discovered that, in advection-dominated regimes, high tortuosity of pore structure promotes homogeneous nucleation by enhancing local mixing through flow disturbance. This model provides a novel framework for the precise regulation of mineral homogeneous nucleation and precipitation, offering critical insights for the optimization of related geological engineering processes.

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I. INTRODUCTION

Mineral nucleation and precipitation ubiquitously exist in geological systems and play a crucial role in many engineering applications. Their dynamic behaviors directly affect the long-term stability and efficiency of the systems. For instance, in the shale gas extraction process, mineral precipitation (e.g., barite and calcium carbonate) may generate crystals/particles and clog micro-fractures or pores within the reservoir, which will significantly reduce the permeability of the matrix as well as the efficiency of the hydrocarbon recovery. As for the geological storage of CO_2, the carbonate precipitation formed by the geochemical reaction of CO_2 with certain minerals can permanently absorb carbon oxide and enhance the long-term safety of storage.[1–7] Moreover, the scaling problem on the heat exchange surface of geothermal systems is also prominent, as mineral precipitation will significantly reduce the heat conduction efficiency and increase the operation and maintenance costs.[8–10] Therefore, a thorough understanding of the nucleation and precipitation processes of minerals as well as the reactive flow in porous structures is of vital significance.

Nucleation of minerals, which is the first step in mineral crystallization, refers to the process by which ions begin to form crystal nuclei from solution.[11–15] This nucleation process plays a crucial role in the subsequent evolution of pore structure and the formation of hydraulic paths in the rock matrix. Previous studies have already made contributions to our understanding of macroscopic mineral growth and precipitation kinetics, whereas the microscale nucleation mechanisms are often neglected by assuming the mineral formed uniformly within the system.[16–18] However, recent advances reveal that the nucleation of minerals, which is typically modeled as a stochastic process, has a significant impact on the patterns of mineral growth and precipitation.[19–23] This probabilistic characteristic of mineral nucleation could lead to a high degree of deviation in the spatial and temporal distribution of precipitated minerals. Moreover, the heterogeneity of nucleation sites can significantly alter the pore structure

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and permeability of porous media, which should be taken into account for reactive flow, especially at the pore scale.[12,24–26]

Typically, the nucleation of minerals occurs either directly within the aqueous phase (homogeneous nucleation) or on the liquid–solid interface of the pore structure (heterogeneous nucleation), depending on the physicochemical conditions of both liquid and solid phases.[27] Currently, most research mainly focuses on the heterogeneous nucleation of mineral crystals due to its lower energy barriers. In contrast, according to the classical nucleation theory (CNT), the homogeneous nucleation in bulk solution typically needs to overcome a much higher energy barrier, which means that its appearance requires either solution to have high supersaturation or additional nucleation sites to be present.[28] Such requirements make the homogeneous nucleation of minerals often neglected in previous studies.

Nevertheless, many recent studies have discovered that the appearance of the homogeneous nucleation of mineral crystals in solution is far more likely to happen than was previously thought. For example, Poonoosamy *et al.*[14] utilized microfluidic droplet technology to generate oversaturated droplet arrays and statistically analyzed the evolution of nucleation probabilities over time through time-lapse optical microscopy. Surprisingly, their experimental measurement showed that the homogeneous nucleation rate of barite crystals in the droplets was much higher than the value predicted by the CNT, which indicated that homogeneous nucleation might not be ignored, especially when the saturation index (SI) of the solution is relatively high. In addition, homogeneous nucleation combined with reactive flow could result in significant pore-blocking effects, which will greatly alter the permeability of the porous matrix by clogging the pore throat. Prasianakis *et al.*[29] revealed the microscopic mechanism of mineral precipitation in porous media by coupling the nucleation of minerals with the lattice Boltzmann method (LBM). Their research found that homogeneous nucleation dominates in highly supersaturated solutions, which would form nanocrystalline precipitates to fill the pores, thereby rapidly altering the permeability and transport properties of the porous media. In a simulation study on the pore-scale dynamics of precipitation in diffusion-dominated porous systems, Deng *et al.*[11] investigated the effect of homogeneous nucleation on mineral precipitation by adding a nucleation correction term to the reaction rate. Even though the minerals formed by homogeneous nucleation are treated as a fixed solid phase in this study, the results showed that homogeneous nucleation within solutions led to the formation of a continuous precipitation band of minerals in the pore space rather than adhering to the surface of mineral grains. This phenomenon is consistent with the mixed-induced precipitation mode observed in microfluidic experiments,[30,31] which further highlights the necessity of deepening the understanding of the mechanism of homogeneous nucleation at the pore scale for precisely regulating mineral precipitation behaviors.

Although these studies have advanced our understanding of homogeneous nucleation in the evolution of the pore structure, the underlying mechanism of how homogeneous nucleation occurs in reactive flow, especially within the porous matrix, remains unclear. To analyze this complex process, the homogeneous nucleation in porous media must be coupled with reactive flow as well as particulate transport, which is not a trivial task. More specifically, after mineral crystals are formed in solution, their movement is regulated by both fluid dynamics and the interaction between crystal particles and matrix, which will eventually affect the structure and physical properties of porous media. Currently, the numerical studies on particle flow at the pore scale have attracted increasing attention, and it has wide applications in industrial manufacturing, and energy- and environmental-related areas.[32,33] More specifically, the Euler–Lagrange method based on the discrete element method (DEM) or discrete particle method (DPM) was used to model the clogging of flow channels with porous structure, while the fluid flow and transport of species were solved by several different modeling approaches such as the lattice Boltzmann method (LBM) or Darcy–Brinkman–Stokes (DBS) method.[34–38]

Currently, a disconnection between micro-scale nucleation and pore-scale reactive transport in porous media still exists, and their interplay largely remains to be explored. This is mainly because the existing models on particle flow in porous media usually simplify the initial homogeneous nucleation process (or simply regard it as a steady solid phase) and focus on the transport of already formed particles. Even in the recent studies on pore-scale mineral precipitation that incorporate the nucleation process, the main focus has been on the heterogeneous nucleation and precipitation phenomena occurring on the surface of the mineral matrix, failing to address the issue of homogeneous nucleation in bulk solution and its subsequent transport.[39,40] To bridge this obvious gap, unlike previous studies, we construct a pore-scale model along with the probabilistic CNT nucleation method and an Euler–Lagrange particle tracking method to simulate the homogeneous nucleation and mineral particle transport at the pore scale. The traditional model usually treats minerals formed by homogeneous nucleation as a steady solid phase, while our model initializes them as discrete crystal particle entities that can flow with fluids. This model is more in line with the actual situation because mineral particles are not static in the real geological environment but constantly transport along with the flow of the fluid (see Fig. 1). By introducing this dynamic description into the model, we can more accurately simulate the behavior of mineral particles in fluids, thereby significantly enhancing the accuracy and reliability of prediction.

This manuscript is organized as follows. First, we elaborate on the numerical methods for the coupled homogeneous nucleation process of minerals and the flow of crystal particles in detail. Then, this newly developed solver is verified against both the analytical solution and experimental measurements to prove its accuracy in modeling the homogeneous nucleation process. Following this, we modeled the influence of different flow conditions and porous structures on the efficiency of homogeneous nucleation. In this way, the interplay of

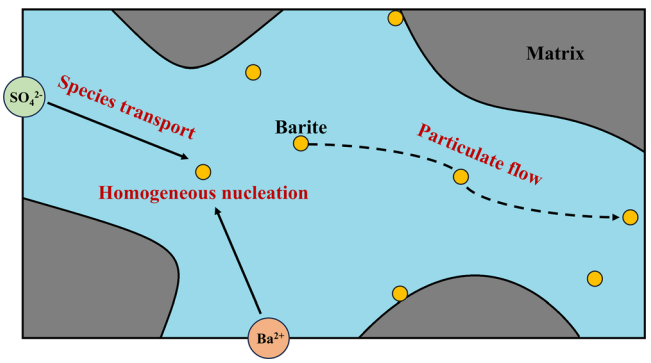


FIG. 1. Schematic diagram of homogeneous nucleation and particulate flow of minerals in porous media.

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nucleation, transport of species, and reactive flow is investigated, and the impact of the key parameters such as Péclet number, porosity, and tortuosity are analyzed to provide insights on regulating such a complex homogeneous nucleation process.

II. METHODS

A. Model for fluid flow in porous media

In our study, the porous medium is assumed to be fully saturated (i.e., no gas phase is present). To model the complex process involving mineral crystallization, precipitation, and crystal particle flow, we adopt the micro-continuum method (MCM).[41–45] This method treats porous media as a micro-continuum system and utilizes volume fractions of fluid ε_f, solid ε_s, and particle ε_p phases to represent their proportions within a control volume

$$\varepsilon_f = \frac{V_f}{V_c}, \quad \varepsilon_s = \frac{V_s}{V_c}, \quad \varepsilon_p = \frac{V_p}{V_c}, \tag{1}$$

where V_f (m^3), V_s (m^3), V_p (m^3), and V_c (m^3) are the volumes of the fluid phase, solid phase, particle phase, and the total volume. For every control volume, the summation of volume fractions of different phases yields

$$\varepsilon_f + \varepsilon_s + \varepsilon_p = 1. \tag{2}$$

In our model, the fluid flow is governed by the modified Darcy–Brinkman–Stokes (DBS) equation, which has been proved effective in modeling the complex flow within the porous structure:[44,46]

$$\frac{1}{\varepsilon_f}\left[\frac{\partial \rho_f \bar{\mathbf{u}}}{\partial t} + \nabla\cdot\left(\frac{\rho_f}{\varepsilon_f}\bar{\mathbf{u}}\bar{\mathbf{u}}\right)\right] = -\nabla \bar{P}_f + \frac{\mu_f}{\varepsilon_f}\nabla^2\bar{\mathbf{u}} - \frac{\mu_f}{K}\bar{\mathbf{u}} + S_p, \tag{3}$$

where ρ_f ($kg\,m^{-3}$) is the fluid density, $\bar{\mathbf{u}}$ ($m\,s^{-1}$) is the superficial velocity of fluid, $\bar{\mathbf{u}} = \frac{1}{V_c}\int_{V_f}\mathbf{u}\,dV$, $\mathbf{u}$($m\,s^{-1}$) is the true velocity of the fluid, $\bar{P}_f$ ($kg\,m^{-1}\,s^{-2}$) is the intrinsic phase averaged pressure, $\bar{P}_f = \frac{1}{V_f}\int_{V_f} P dV$, P ($kg\,m^{-1}\,s^{-2}$) is the pressure of the fluid, μ_f ($kg\,m^{-1}\,s^{-1}$) is the dynamic viscosity, and K (m^2) is the local permeability, which can be expressed in the form of a certain function of porosity, such as the Kozeny–Carman equation

$$K = K_0\frac{\varepsilon_f^3}{(1-\varepsilon_f)^2}, \tag{4}$$

where K_0 is the Kozeny–Carman constant for certain porous structure.

The two-way coupling between the fluid phase and the particle phase is achieved via including a source term S_p in the momentum equation. As the fluid drag acting on each particle is calculated, according to Newton's third law of motion, the source term is computed as a volumetric fluid–particle interaction force given by[32,38,47,48]

$$S_p = -\frac{\sum_{i=1}^{N_p}\mathbf{F}_{drag}}{V_c}, \tag{5}$$

where F_{drag} (N) is the viscous drag force acting on the crystal particles, V_c is the volume of a computational cell, and N_p is the number of particles located in this cell.

B. Model for particle dynamics

In the current study, as we are focusing more on the homogeneous nucleation process, both the rotational effects of crystal particles and the interaction forces between particles are neglected for the sake of computational efficiency. Hence, the governing equations for the particle phase are simplified as follows:[38,47]

$$\frac{\partial \mathbf{x}_p}{\partial t} = \mathbf{u}_p, \tag{6}$$

$$m_p\frac{\partial \mathbf{u}_p}{\partial t} = \mathbf{F}_{drag} + \mathbf{F}_G, \tag{7}$$

where $\mathbf{x}_p$ (m) is the position vector of the particle, $\mathbf{u}_p$ ($m\,s^{-1}$) is the particle velocity, m_p (kg) is the particle mass, and $\mathbf{F}_{drag}$ and $\mathbf{F}_G$ are the viscous drag force and the effective gravitational force, respectively, acting on the particle, which are expressed as

$$\mathbf{F}_{drag} = \beta\frac{V_{particle}}{\varepsilon_p}(\mathbf{u}_f - \mathbf{u}_p), \tag{8}$$

$$\mathbf{F}_G = V_{particle}(\rho_p - \rho_f)\mathbf{g}, \tag{9}$$

where $V_{particle}$ (m^3) is the particle volume for spherical particle $V_{particle} = \frac{\pi d_p^3}{6}$ and d_p (m) is the diameter of the crystal particle, $\mathbf{u}_f$ ($m\,s^{-1}$) is the fluid velocity, ρ_p ($kg\,m^{-3}$) is the particle density, $\mathbf{g}$ ($m\,s^{-2}$) is the gravity acceleration, β is the inter-phase momentum exchange coefficient for dilute suspensions, which is given by the widely used Wen–Yu law (it provides an empirical drag law for particles in fluid flow as a function of Reynolds number and the particle phase volume fraction) in this study:[47,49]

$$\beta = \frac{3}{4}C_D\frac{\varepsilon_f\,\varepsilon_p}{d_p}\rho_f\,|\mathbf{u}_f - \mathbf{u}_p|\varepsilon_f^{-2.65}, \tag{10}$$

where C_D is the drag coefficient, it is defined as a function of the particle Reynolds number Re_p, and given by the Schiller–Naumann empirical correlation:

$$C_D = \begin{cases}\dfrac{24}{Re_p}\left(1 + 0.15Re_p^{0.687}\right), & Re_p \le 1000,\\ 0.44, & Re_p > 1000,\end{cases} \tag{11}$$

$$Re_p = \frac{\rho_f\,|\mathbf{u}_f - \mathbf{u}_p|d_p}{\mu_f}. \tag{12}$$

C. Model for transport of species

To capture the spatiotemporal evolution of species at the pore scale, the transport of multiple solute components is described by the advection–diffusion–reaction equation:[22,40]

$$\frac{\partial \varepsilon_f C_i}{\partial t} + \nabla\cdot(\bar{\mathbf{u}}C_i) = \nabla\cdot\left(D_{eff}\nabla C_i\right) - \dot{R}_n, \tag{13}$$

where C_i is the concentration of solute component i, D_{eff} ($m^2\,s^{-1}$) is the effective diffusion coefficient in the form of $D_{eff} = \varepsilon_f^2 D$ and D ($m^2\,s^{-1}$) is the bulk diffusion coefficient. $\dot{R}_n$ ($mol\,m^{-3}\,s^{-1}$) is the source term related to nucleation rate, which is also related to the volume fraction by

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$$\frac{\partial \varepsilon_p \rho_p}{\partial t} = M_p \dot{R}_n, \tag{14}$$

where M_p (kg mol^{-1}) is the molar mass for the particle phase. $\dot{R}_n$ is associated with the probability function ($\mathbf{\Gamma}_c$) of mineral nucleation within a unit volume for a given nucleation period t_{unit} (s):[39]

$$\dot{R}_n = \frac{M_{nucleus}\mathbf{\Gamma}_c}{V_c t_{unit}}, \tag{15}$$

$$M_{nucleus} = \frac{\rho_p V_{particle}}{M_p}, \tag{16}$$

where $M_{nucleus}$ (mol nucleus^{-1}) is the mole number per nucleus, $\mathbf{\Gamma}_c$ (nuclei) is the number of nuclei formed in the solution within a computational volume cell during t_{unit}. The comprehensive details of Eq. (15) can be found in previous work.[39] $\mathbf{\Gamma}_c$ was calculated by the following relation:

$$\mathbf{\Gamma}_c = J_c \times V_f \times t_{unit}, \tag{17}$$

where J_c (nuclei m^{-3} s^{-1}) is the local nucleation rate within the computational volume cell, V_f (m^3) is the volume of the fluid within the computational volume cell, and t_{unit} (s) is the nucleation period, which is set as 1 s for this study. The homogeneous nucleation rate can be computed following the classical nucleation theory (CNT):[28]

$$J = A \exp\left(-\frac{\Delta G_c}{kT}\right), \tag{18}$$

$$\Delta G_c = \frac{b\nu_0^2\sigma^3}{(kT\ln\Omega)^2}, \tag{19}$$

where J (nuclei m^{-3} s^{-1}) is the homogeneous nucleation rate, A (nuclei m^{-3} s^{-1}) is a pre-exponential factor, k (J K^{-1}) is the Boltzmann constant, T (K) is the temperature of the solution, ΔG_c (J) is the energy barrier required for the formation of a nucleus of critical size, b is a geometry factor that depends on the shape of the nucleus, ν_0 (m^3) is the molecular volume of the particle phase, and σ (J m^{-2}) is the interfacial tension. Ω is the saturation ratio, with the saturation index (SI) defined as SI $= \log_{10}\Omega$. For barite, we have $\Omega = \frac{[Ba^{2+}][SO_4^{2-}]}{K_{sp}}$, where K_{sp} is the solubility product constant, which is obtained from the PHREEQC database.

D. Model for stochastic homogeneous nucleation

In the multi-scale simulation of the crystallization and precipitation, the randomness of homogeneous nucleation events is usually described by a stochastic method. In this study, we introduce a nucleation indicator function to characterize whether nucleation occurs at a specific location (e.g., in a computational volume cell) within a given period. The definition of this function is based on the following probability model. Let the random variable ξ follow a uniform distribution over the interval [0,1], i.e., $\xi \sim U(0,1)$. We define the nucleation indicator function I_{nuc} as

$$I_{nuc} = \begin{cases} 1, & \mathbf{\Gamma}_c > \xi \\ 0, & \mathbf{\Gamma}_c \le \xi \end{cases}. \tag{20}$$

The homogeneous nucleation event in each computational volume cell is governed by the probability function $\mathbf{\Gamma}_c$[see Eq. (17)] depending on the fluid volume V_f (m^3) and the local nucleation rate J_c (nuclei m^{-3} s^{-1}), with $\mathbf{\Gamma}_c$ applicable only when V_f is sufficiently small during the nucleation period and to prevent multiple nucleation events within a volume cell. We assume nucleation occurs exclusively in the bulk fluid phase ($\varepsilon_f > 0.99$), excluding nucleation on substrate surfaces or existing crystals. To calculate $\mathbf{\Gamma}_c$ and determine whether a nucleation event occurs, a uniformly distributed random number ξ is generated using the standard C++ math library RNG (Random Number Generator). The nucleation probability was calculated for each computational cell at the beginning of each period, t_{unit}. In this study, we set it to be $t_{unit} = 1$ s. If the condition $\mathbf{\Gamma}_c > \xi$ is satisfied, the cell is marked as nucleated with $I_{nuc} = 1$; otherwise, no nucleation is triggered and $I_{nuc} = 0$.The algorithm iterates through all computational cells in the domain, performing independent nucleation checks to generate spatially random nucleation sites that maintain physical consistency. Upon selection of a computational cell for nucleation, a spherical particle is initialized within the cell volume. This model is implemented in the RPFFoam (ReactiveParticulateFlowFoam) solver using the open-source CFD package OpenFOAM.[50] The numerical solver is developed using the C++ programming language and realizes the automatic management and post-processing of cases through Bash scripts and Python. The entire computing process runs on the Linux operating system environment and applies a parallel computing strategy based on domain decomposition, thereby significantly improving the solution efficiency on the computing cluster.

III. MODEL VERIFICATION

A. Verification of the numerical model

To verify the efficiency of the RPFFoam solver on modeling homogeneous nucleation, we constructed a system consisting of oversaturated barium sulfate solutions going through homogeneous nucleation processes. Here, as we focus on the verification of the nucleation process, the solute does not transform into a solid phase during the nucleation process to prevent interference from phase change. The simulation domain was set as a three-dimensional cube, as shown in Fig. 2(a), with dimensions of $10 \times 10 \times 10$ mm^3. Spherical nuclei [the blue particles in Fig. 2(a)] represent the occurrences of individual homogeneous nucleation events in the solution. The whole simulation domain is maintained at a fixed ion concentration of $C_1 = C_2 = 0.261$ for Ba^{2+} and SO_4^{2-}, which corresponds to a saturation index SI = 2.8. Other parameters in this simulation related to the nucleation were referred to the microfluidic experiments on barite nucleation conducted by Poonoosamy *et al.*,[14] which were also listed in Table I. Note the pre-exponential factor A is fitted from the experimental data according to Poonoosamy's measurements.

The total nuclei and nucleation rate results from ten sets of individual simulations are averaged and plotted in Figs. 2(b) and 2(c) to mitigate the stochastic nature of probabilistic nucleation. The black dashed lines represent the prediction of crystal nuclei calculated from CNT using Poonoosamy's measurements.[14] The error bar shows the fluctuation range of the simulated nucleation quantities. To prevent the numerical errors introduced from mesh quality, we also conducted mesh size analysis with the homogeneous nucleation model. Figure 2(d) shows the relationship between the simulation results of nucleation rates (red rhombus points) and Poonoosamy's results (black dashed line) with different discretizing sizes. As shown by the verification simulation, there is a high degree of consistency between our

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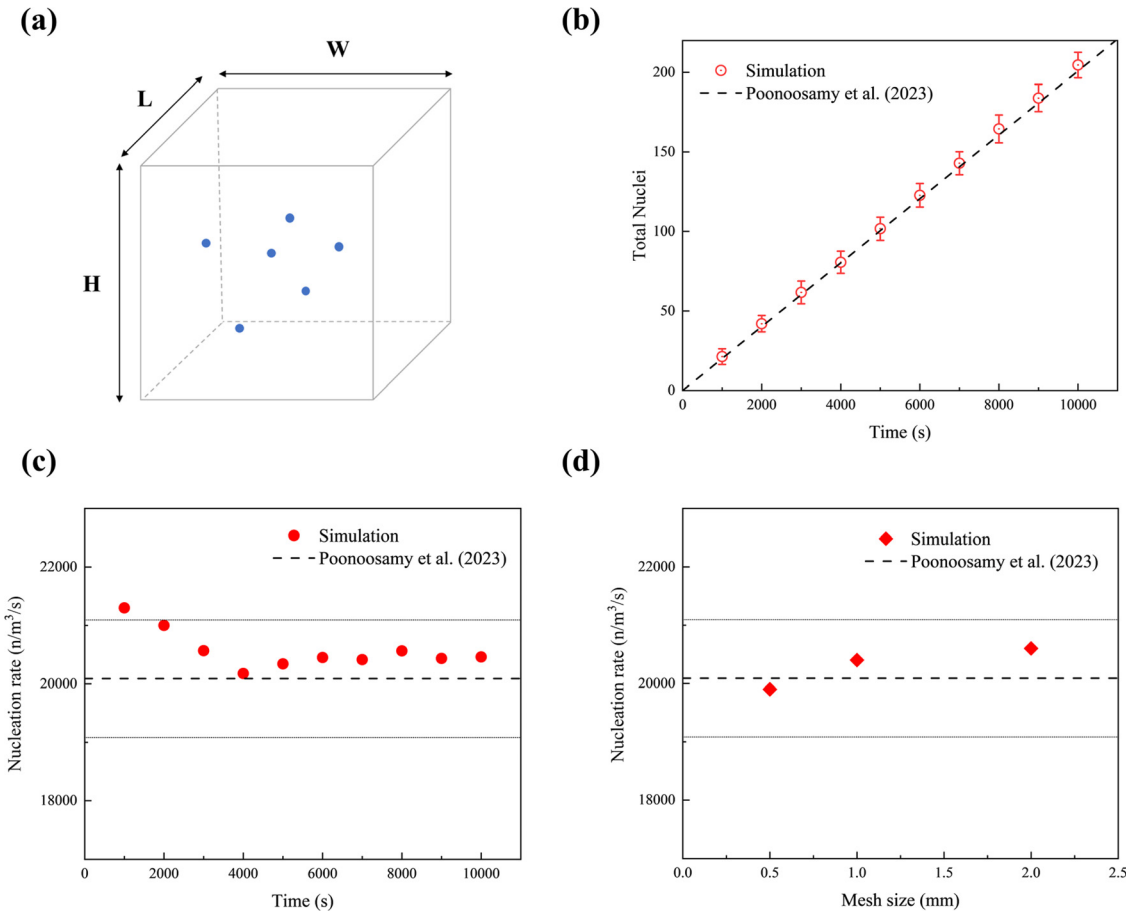


FIG. 2. Validation of homogeneous nucleation model against experimental data. (a) Model schematic diagram. The blue dots represent spherical crystal particles. (b) Comparison of the simulation and analytical solutions of the total number of crystal nuclei varying with time. The red circles represent the total number of crystal nuclei, and the black dashed line represents the calculated number of crystal nuclei from experimental data. The error bar indicates the fluctuation range of the data. (c) The nucleation rate varies with time. The solid red circles represent the average nucleation rate. The thin black dash lines represent the 5% error line. (d) The relationship between nucleation rate statistics and cell size.

simulation results and previous experimental work, and all the simulation results are within the allowable error range, while the nucleation model is not sensitive to mesh sizes. Hence, the RPFFoam solver developed in this study could accurately model the homogeneous nucleation of barite crystals in oversaturated solutions.

B. Verification of the particle dynamics

In addition to verifying the homogeneous nucleation module, to demonstrate the RPFFoam's ability to couple the reactive flow and nucleated crystal motion in complex dynamic scenarios, we conducted the fluid–particle interaction simulations using the current model.

1. The sedimentation of a particle in solution

The inset in Fig. 3(a) shows the schematic diagram of the simulation system, which predicts the movement of a single particle settling in bulk solution due to gravity. The simulation domain is a two-dimensional rectangular box with dimensions set at $10 \times 10\,\mathrm{mm}^2$ and a mesh size of 0.1 mm. The fluid and particle remain stationary within the simulation domain at the start of the simulation. Due to the fluid–particle interaction, the particle motion is driven by the competition between gravity and drag force from fluid. It will accelerate until these forces reach equilibrium, and the particle will reach its terminal velocity. We verify the entire process of particle sedimentation movement by comparing the simulated particle velocity profile with the Stokes' theoretical solution.[51] The expression of the theoretical sedimentation velocity $v_1(t)$ $(\mathrm{m\,s^{-1}})$ is given as follows:

$$v_1(t) = v_t\left(1 - e^{-t/\tau_p}\right), \tag{21}$$

$$v_t = \frac{(\rho_p - \rho_f)g d_p^2}{18\mu_f}, \tag{22}$$

$$\tau_p = \frac{\rho_p d_p^2}{18\mu_f}, \tag{23}$$

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TABLE I. Modeling parameters in the verification simulations.

Modeling parameters	Value	Units
Section III A		
A	$3.19 \times 10^{25}\,N_A$	nuclei $m^{-3}\,s^{-1}$
b	$16\pi/3$	
v_0	8.6×10^{-29}	m^3
σ	0.134	$J\,m^{-2}$
T	298.15	K
K_{sp}	$1.096\,48 \times 10^{-10}$	
$C_1 = [SO_4^{2-}]$	0.261	mol m^{-3}
$C_2 = [Ba^{2+}]$	0.261	mol m^{-3}
Section III B 1		
ρ_p	2000	kg m^{-3}
d_p	2×10^{-5}	m
ρ_f	1000	kg m^{-3}
μ_f	10^{-3}	kg $m^{-1}\,s^{-1}$
Section III B 2		
U_f	10^{-4}	m s^{-1}
ρ_p	2000	kg m^{-3}
d_p	2×10^{-5}	m
ρ_f	1000	kg m^{-3}
μ_f	10^{-3}	kg $m^{-1}\,s^{-1}$

where v_t (m s^{-1}) is the terminal velocity of the particle and τ_p (s) is the relaxation time of the particle. Here, for the convenience of result analysis, we normalize Eq. (21) by setting the normalized velocity as $v_1^* = \frac{v_1(t)}{v_t}$ and normalized time as $t^* = \frac{t}{\tau_p}$. Then, the theoretical velocity for particle sedimentation is simplified and given as

$$v_1^* = 1 - e^{-t^*}. \tag{24}$$

The specific parameters of the simulation are shown in Table I. Under this condition, we have the theoretical final velocity as $v_t = 2.18 \times 10^{-4}$ m/s and the theoretical relaxation time $\tau_p = 4.44 \times 10^{-5}$ s. Figure 3(a) shows the comparison of the temporal evolution of dimensionless particle velocity v_1^* between simulation and theoretical prediction, which demonstrates the effectiveness of RPFFoam in calculating the fluid–particle bidirectional coupling mechanism.

2. The dynamics of a particle in a steady stream

The modeling system of the particle motion in a constant stream is similar to the model described in Sec. III C, while the driving force switches from gravity to the drag force from the steady stream flow. The schematic diagram is shown in the inset of Fig. 3(b). The fluid flows through the domain from left to right at a fixed speed. In this simulation, we ignored the influence of particle gravity. The particle was only subjected to drag force and began to accelerate until it reached the same speed as the fluid. We verify the entire process of particle movement in a constant flow field by comparing the simulated output particle velocity with the Stokes' theoretical solution. The theoretical prediction of the particle velocity in a steady flow field follows the expression:

$$v_2(t) = U_f\left(1 - e^{-t/\tau_p}\right), \tag{25}$$

where U_f (m s^{-1}) is the fluid velocity, τ_p (s) has the same expression as Eq. (23). Similarly, for the convenience of result analysis, we normalize Eq. (25) by setting the normalized velocity as $v_2^* = \frac{v_2(t)}{U_f}$ and time $t^* = \frac{t}{\tau_p}$. Then, the formula for the velocity of a particle moving in a steady flow field is rewritten as

$$v_2^* = 1 - e^{-t^*}. \tag{26}$$

The specific parameters are shown in Table I, and the theoretical relaxation time is obtained as $\tau_p = 4.44 \times 10^{-5}$ s. As shown in Fig. 3(b), the simulated particle velocity (red dots) matches the theoretical solution (blue solid line), which indicates that the temporal evolution of particle velocity obtained from our model is consistent with the prediction of the analytical solution. According to the above

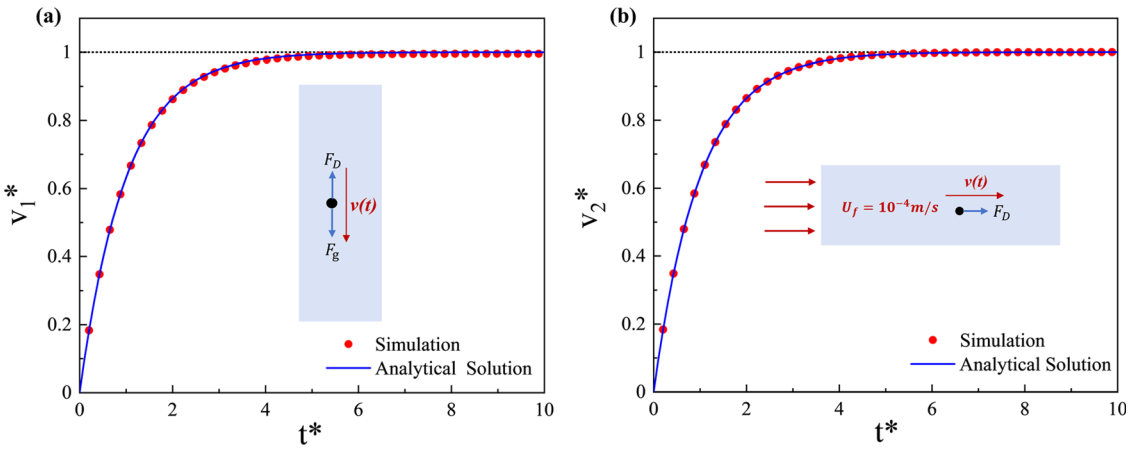


FIG. 3. (a) The curve of characteristic velocity v_1^* varying with characteristic time t^*. The inset shows a schematic diagram of the sedimentation motion of a single particle (rectangular domain, with dimensions of $10 \times 10\,mm^2$). (b) The variation curve of the characteristic velocity v_2^* with the characteristic time t^*. The inset shows a schematic diagram of a single particle moving in a constant flow field.

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verification results, we believe the RPFFoam possesses the ability to model the coupled fluid and particle motion in complex dynamic scenarios.

C. Overview of the modeling systems

In this study, we have constructed two types of simulation systems to investigate the interplay between the transport of species and the homogeneous nucleation of minerals in porous media. As shown in Fig. 4(a), the first type of simulation system is to mimic the mixing and nucleation process in a microchannel. The 2D simulation domain is a three-dimensional rectangular microchannel with dimensions of $1000 \times 100\ \mu m^2$, which has a characteristic length of $H = 100\ \mu m$.

Other than the microchannel, a matrix system is also constructed by inserting cylindrical solids with different sizes into the channel to tune the porosity of the system [see Fig. 4(b)]. By setting the top and bottom boundaries as symmetrical, it allows us to study the homogeneous nucleation within the porous matrix. Although this system is a very simplified representation of a porous medium, it has all the fundamental features of a porous material, i.e., it contains several representative elementary volumes (REV) with porosity, permeability, and tortuosity. This idealized porous medium is also commonly used to investigate complex flow and transport in porous media from the pore scale and to get more insights regarding the different mechanisms involved.[46,52] To make the governing equations described below valid all over the computational domain, i.e., in the free and the solid regions, we assume that the cells describing cylindrical solid regions always contain a tiny porosity, $\varepsilon_f \equiv 0.001$. In the porous matrix system, the diameter of the cylinder is represented by d, and the distance from the center of the cylinder to the centerline of the system is set as X.

In all systems, the ion concentrations C_1 and C_2 are injected from the upper left inlet and the lower left inlet, respectively. To facilitate analysis of the effects of advection–diffusion on the homogeneous nucleation rate of barite in solution and the distribution of ion concentration, we define the normalized nucleation rate as $J^* = \frac{J}{J_{max}}$ and the normalized concentration as $C^* = \frac{C}{C_{in}}$. Here, J_{max} is the maximum nucleation rate obtained from Eq. (18) using inlet ion concentration C_{in} by assuming both species are well mixed. Hence, J^* and C^* varied within the range from 0 to 1 for a system undergoing nucleation and precipitation.

As the solutions were injected into these systems, the species started to mix, and the homogeneous nucleation would occur. According to the classical nucleation theory (CNT), we can know that the energy barrier required for heterogeneous nucleation is significantly lower than that for homogeneous nucleation. This means that in the actual geological engineering process, when the saturation index is low, the system is more prone to heterogeneous nucleation. Meanwhile, studies have shown that in relatively small pores, the homogeneous nucleation process is inhibited, which in turn makes heterogeneous nucleation dominant. From the perspective of the entire geological evolution process, in the early stage of the reaction or when the saturation has not yet reached the threshold that triggers homogeneous nucleation, heterogeneous nucleation plays a significant role as the main precipitation mechanism.[29] However, with the passage of time and changes in local environmental conditions, the main nucleation mechanisms may change, and this dynamic change fully demonstrates the complexity of the evolution of geological systems. In this study, to avoid the complexity of heterogeneous nucleation, we assume the solid phase of porous structure is extremely unfavored for heterogeneous nucleation of minerals. In this way, we could focus on analyzing the homogeneous nucleation process more clearly. Figure 4(c) shows the distribution of crystal particles in microchannels under the condition of $Pe = 1$, and Fig. 4(d) shows the distribution in the porous matrix under the same condition. For better visualization, the sizes of the nucleated crystal particles are magnified by 15 times.

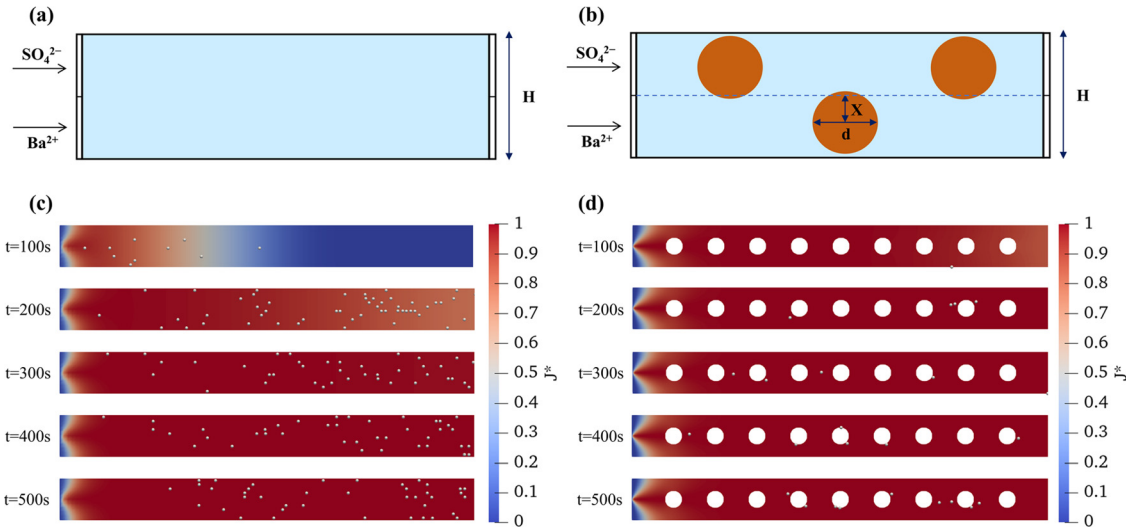


FIG. 4. (a) Schematic diagram of the microchannel model. (b) Schematic diagram of the ideal porous medium model. (c) and (d) The distribution of crystal particles in microchannels and porous media at different times (100, 200, 300, 400, and 500 s).

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IV. RESULTS AND DISCUSSION

In this section, we present all results from the simulation in order: first, we simulated the influence of microchannels and porous media under different *Pe* numbers on the homogeneous nucleation; then, we explored the effects of porosity and tortuosity on the homogeneous nucleation of minerals. Modeling parameters used in these simulations are listed in Table II.

A. Effect of Péclet number on homogeneous nucleation

To explore the influence of transport mechanisms on the homogeneous nucleation behaviors of barite, we investigated nucleation with different Péclet numbers ($Pe = \frac{UH}{D}$) by tuning the fluid inlet velocity U, which, in turn, adjusts the competition between advection and diffusion of species. Figures 5(a)–5(d) show the spatial distributions of C_1^* and C_2^* within the microchannels at t = 500 s and t = 1000 s for different *Pe* numbers. As the C_2^* distribution is symmetrical against C_1^* over the X–Y plane at the centerline of the simulation domain, therefore, we only analyze C_1^* in this study.

When the *Pe* numbers are relatively low ($Pe = 0.01$ and $Pe = 0.1$), the diffusion of species remains the dominant transport mechanism, and the ions are well mixed near the inlet of the channel. However, their transport toward the right-side outlet was limited by the diffusion process. Hence, the concentration near the outlet is much lower than that near the inlet, which leads to a lower nucleation rate, especially at 500 s. The homogeneous nucleation was suppressed by the low transport efficiency from diffusion at early stages of simulation. This disequilibrium of ion concentration gradually fades as diffusion transport persists, which could be quantified by the diffusion length scale, $\sqrt{Dt}$. At the moderate *Pe* number ($Pe = 1$), both contributions from advection and diffusion on the transport of species became important, which enhanced the mixing process and led to a more uniform distribution of concentration. As the *Pe* number further increases, under the conditions of high *Pe* numbers ($Pe = 10$ and $Pe = 100$), the advection effect becomes more significant and resulted in a rapid increase in the concentration gradient in the channel along the y axis. At this point, a sharp concentration distribution is formed in the flow direction. By comparing the distribution of C_1^* under t = 500 s and t = 1000 s, the concentration field reached steady state within a short period of time in the high *Pe* case, while it required a longer time for the low *Pe* case.

Based on the ion concentration field, we could further analyze the evolution of the homogeneous nucleation rate in detail. Figures 5(e) and 5(f) show the distribution of J^* in the microchannels at different *Pe* numbers for t = 500 and 1000 s. It was observed that, in the low *Pe* number cases, the nucleation rate varied as the ions underwent the mixing process, which showed a clear moving front due to diffusion. In contrast, in the high *Pe* number cases ($Pe = 100$), the nucleation rate distribution is clearly separated into two regions. The higher nucleation rate appeared within a stable cone region, while the lower nucleation rate occupied the upper and lower triangular-shaped regions near the inlet. This distinguished deviation of nucleation rate in different regions could be attributed to the fast advection

TABLE II. Modeling parameters in the homogeneous nucleation simulations.

Modeling parameters	Value	Units
Section III A		
A	2.905×10^{13}	nuclei $m^{-3}\,s^{-1}$
b	$16\pi/3$	
v_0	8.6×10^{-29}	m^3
σ	0.134	$J\,m^{-2}$
T	298.15	K
K_{sp}	$1.096\,48 \times 10^{-10}$	
$C_1 = [SO_4^{2-}]$	0.2	$mol\,m^{-3}$
$C_2 = [Ba^{2+}]$	0.2	$mol\,m^{-3}$
D	1.0×10^{-9}	$m^2\,s^{-1}$
H	100	μm
Section IV A		
U	1×10^{-7}, 1×10^{-6}, 1×10^{-5}, 1×10^{-4}, 1×10^{-3}	$m\,s^{-1}$
Section IV B		
X	0	μm
U	1×10^{-6}, 1×10^{-5}, 1×10^{-4}, 1×10^{-3}	$m\,s^{-1}$
d	20, 40, 60, 80	μm
Section IV C		
X	0, 5, 10, 15, 20	μm
U	1×10^{-6}, 1×10^{-5}, 1×10^{-4}, 1×10^{-3}	$m\,s^{-1}$
d	40	μm

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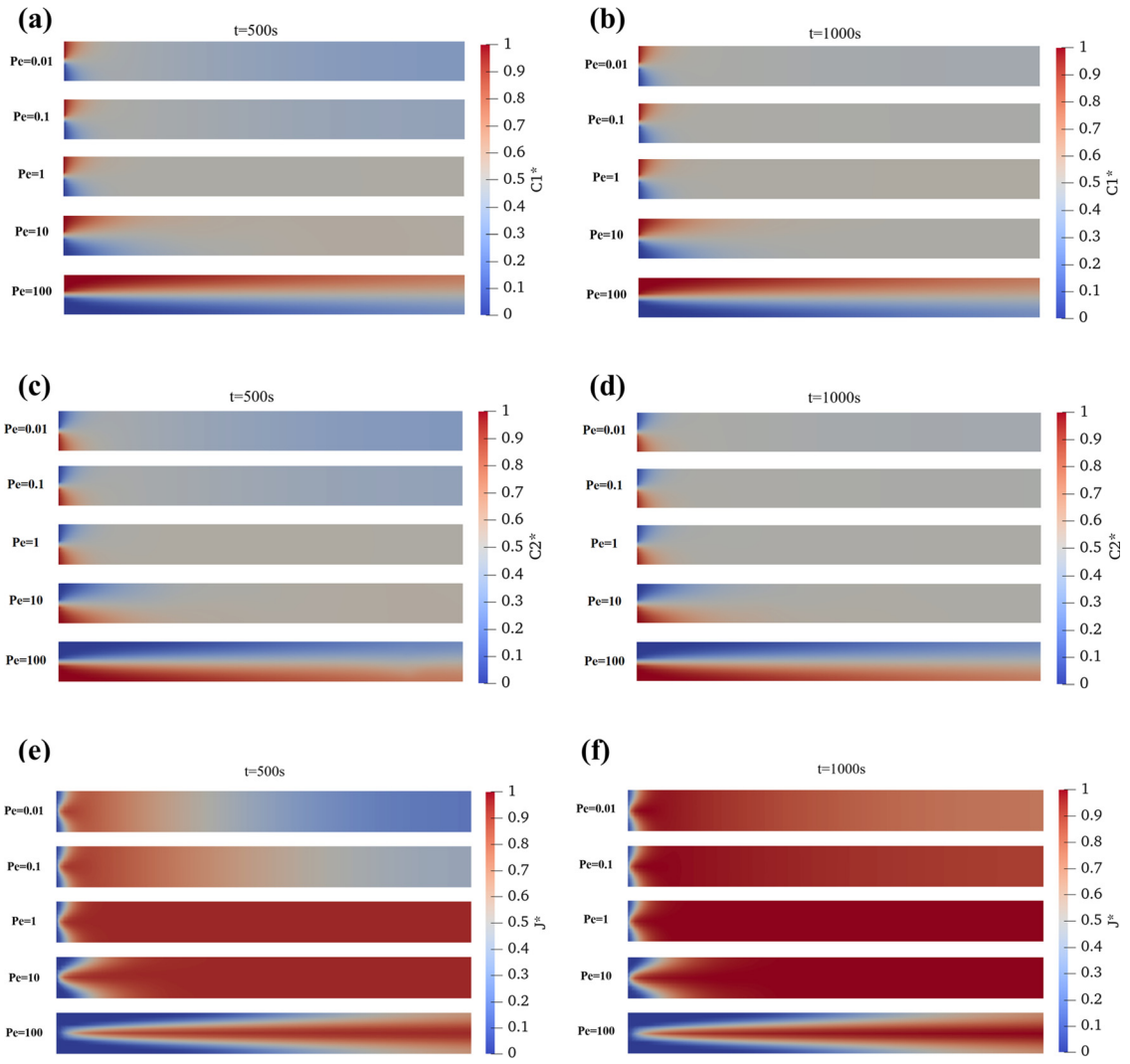


FIG. 5. (a), (c), and (e) The distribution of C_1^*, C_2^*, and J^* at different Pe numbers (the cases of $Pe = 0.01$, $Pe = 0.1$, $Pe = 1$, $Pe = 10$, and $Pe = 100$) when t = 500 s. All simulation snapshots were taken on the X–Z plane at the centerline of the simulation domain at t = 500 s. (b), (d), and (f) The distribution of C_1^*, C_2^*, and J^* at different Pe (0.01, 0.1, 1, 10, 100) numbers when t = 1000 s. All simulation snapshots were taken on the X–Z plane at the centerline of the simulation domain at t = 1000 s.

transport of species, which resulted in the species leaving the simulation domain without being fully mixed. As the nucleation rate is a function of two combined ion concentrations, the high nucleation rate can only exist in the cone-shaped region near the centerline, in which the mixing process could happen through limited diffusion and fluid flow. Unlike these two scenarios, when the Pe number is moderate ($Pe = 1$), the distribution of the nucleation rate J^* is relatively uniform due to the combined contribution from advection and diffusion, and the average nucleation rate within the entire channel is the highest. Under such conditions, it is more favorable for the homogeneous nucleation of barite to happen.

The statistical results on the homogeneous nucleation occurrence further supported this observation. Figure 6(a) shows the variation trend of the total number of crystal nuclei in the microchannel with the Pe number at different times. We found that, with the increase in the Pe number, the trend of the total number of nuclei in the channel at different times was first increasing and then decreasing. Figure 6(b) shows the evolution law of the average nucleation rate k ($\mathrm{n\,s^{-1}}$) within the channel over time under $Pe = 1$ and $Pe = 10$. For each computational volume cell, $k_c = J_c \times V_c$. In the initial stage (t < 3000 s), the average nucleation rate under $Pe = 10$ was significantly higher than that under $Pe = 1$. However, after 3000 s, the average nucleation rate was surpassed by $Pe = 1$. This observation is because, at $Pe = 10$, advection was more dominant, and the solute was rapidly transported into the channel, significantly increasing the average nucleation rate in the short term (t < 3000 s). At $Pe = 1$, both diffusion and

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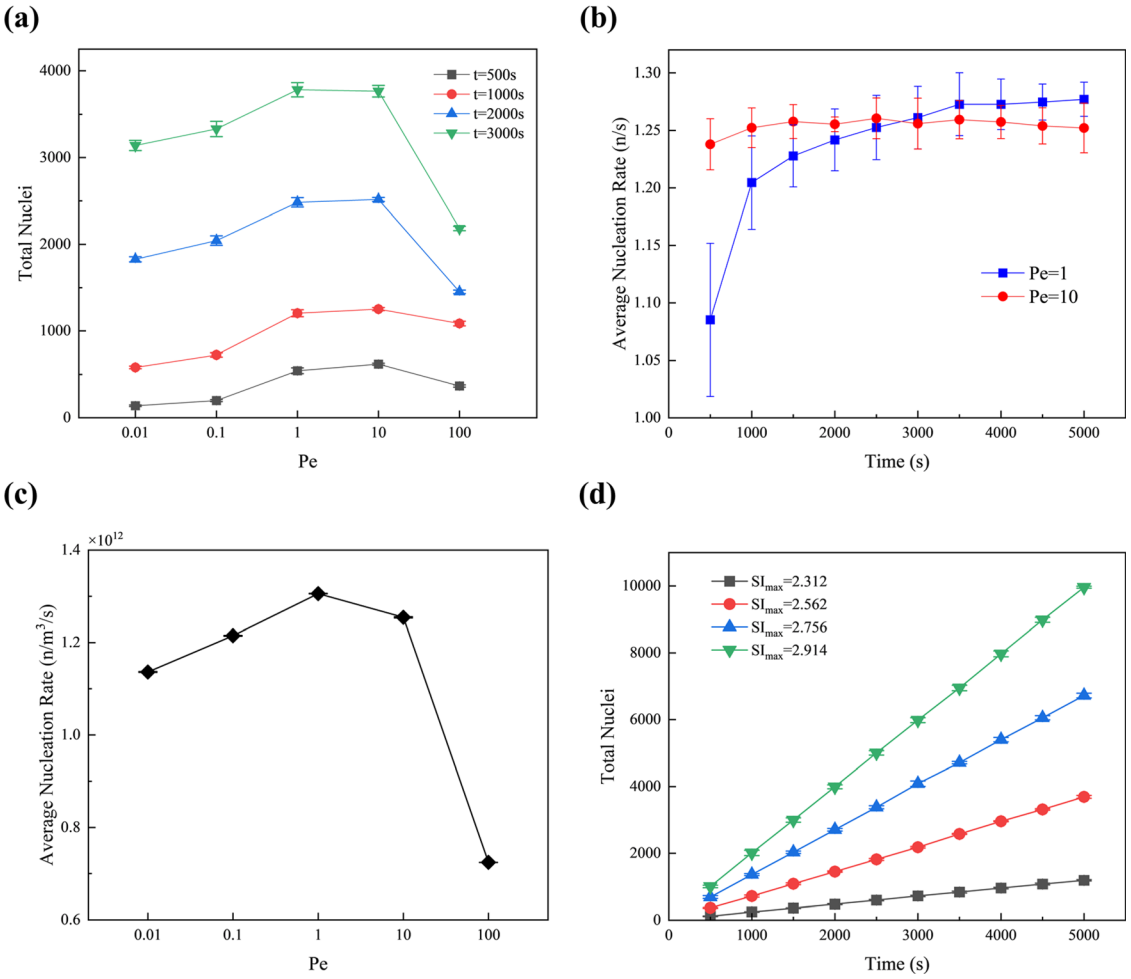


FIG. 6. (a) The evolution of the total nuclei in microchannels as a function of *Pe* number at different times. (b) The evolution of the effective nucleation rate in microchannels over time. (c) The average nucleation rate within the channel under different *Pe* numbers at t = 3000 s. (d) The total number of nuclei vs time at different SI_{max} cases.

advection mechanisms were significant, which maintained the stability of local concentration gradients and avoided premature depletion of solutes. This situation achieved a higher long-term (t > 3000 s) nucleation efficiency, which was more conducive to mineral homogeneous nucleation. Figure 6(c) shows the average nucleation rate within the channel under different *Pe* numbers. We found that, when *Pe* fell into the range O(1–10), the average nucleation rate within the channel was the highest. In this regime, advection and diffusion reached a balance, and the average nucleation rate within the channel is the highest. Thereby, it significantly promoted homogeneous nucleation of minerals within the microchannel. Furthermore, further exploration revealed that under high *Pe* numbers, advection dominated and allowed ions to be transported rapidly. This intense advection disrupts the local mixing process required for nucleation, preventing the formation of local supersaturated regions, which in turn reduces the average nucleation rate and hinders the homogeneous nucleation.

We further explored the influence of the saturation index on homogeneous nucleation within the microchannel at *Pe* = 100 by altering the ion concentrations C_1 and C_2. Figure 6(d) shows the total number of nuclei vs time at different SI_{max} cases. Here, SI_{max} is the maximum saturation index calculated by assuming that both species are well mixed and using the inlet ion concentrations C_1 and C_2. It was observed that the higher the SI_{max}, the steeper the curve rises, which indicates that the homogeneous nucleation rate is faster. We also found that, as the SI_{max} increased, the total number of nuclei in the microchannel also increased significantly. It indicates that under fixed fluid flow conditions, increasing SI can simultaneously significantly enhance both the homogeneous nucleation rate and the total nuclei within the microchannel, thereby comprehensively strengthening the homogeneous nucleation process.

The above analysis demonstrates that an optimal *Pe* number range (in this case, *Pe* = 1–10) exists, which enables the system to reach the highest homogeneous nucleation rate, thus maximizing nucleation yield in the long term. Deviation from this optimal range

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(whether the diffusion is too slow or the convection is too strong) could lead to a decrease in overall nucleation efficiency for a microchannel or single-pore system.

B. Effect of porosity on homogeneous nucleation

To explore the influence of porosity on the homogeneous nucleation of minerals under an advection-dominant regime, we fixed the fluid inlet velocity $U = 1 \times 10^{-3}\,\mathrm{m\,s^{-1}}$ ($Pe = 100$) and kept the position of the cylindrical solid unchanged. The porosity of the porous medium was varied by adjusting the diameter d of the cylindrical solid, with specific parameters provided in Table II. Figure 7(a) presents the distribution of the dimensionless concentration C_1^* in different porous matrices at t = 50 s. We found that when d gradually increased, it could be clearly observed that the blockage effect from the cylinders on the transport of species, especially in the y axis. Lower porosity led to the obstruction of transport and reduction of the mixing of species. In contrast, the transport of species was more similar to the microchannel case in the higher porosity cases, in which less suppression of mixing was observed. For the lower porosity cases, because of the obvious separation of species, the high nucleation rate J^* only exists near the centerline of the domain [see Fig. 7(b)], which limits the overall occurrence of homogeneous nucleation. Meanwhile, as the porosity increased, the homogeneous nucleation rate field changed from a highly local distribution to a relatively uniform one.

Figure 7(c) plots the temporal evolution of the total number of nuclei for different porosity of the porous structure. It was found that, as the porosity increases, the more nuclei are formed by the homogeneous nucleation in the simulation domain, which is consistent with the variation of the nucleation rate. This trend could also be found by calculating the average nucleation rate within the whole domain [see Fig. 7(d)]. As the results show, the system with higher porosity had a much higher average nucleation rate than the lower porosity case. Overall, within the advection-dominated regime, the high porosity enhances the homogeneous nucleation behavior of minerals in a porous matrix.

This perturbation effect from porosity on the nucleation rate distribution weakens as the Pe number reduces. Figure 8 shows the simulation results with similar settings other than fluid inlet velocity set as $U = 1 \times 10^{-5}\,\mathrm{m\,s^{-1}}$, which corresponds to $Pe = 1$. Figures 8(a) and 8(b) present the normalized concentration C_1^* at t = 100 s and t = 1000 s for different porosity. At t = 100 s, solute distribution is relatively uniform in the case with larger cylinders ($d = 80\ \mu$m), while a pronounced stagnant zone (indicated in deep

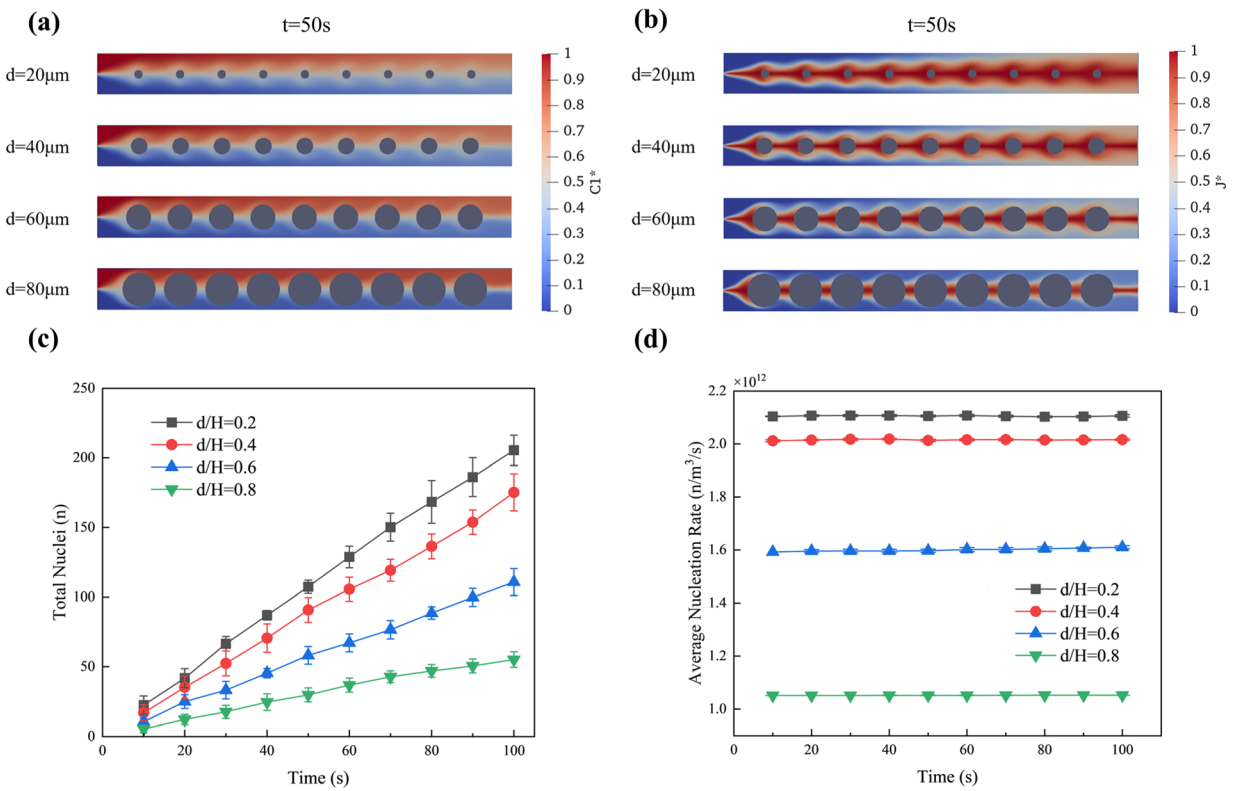


FIG. 7. (a) The distribution of C_1^* at different porosity at t = 50 s. Nine cylindrical solids are uniformly arranged along the X–Z central axis of the simulation domain. The four systems, with diameters of 20, 40, 60, and 80 μm, have porosity of 0.972, 0.887, 0.746, and 0.548, respectively. All simulation snapshots were taken on the X–Z plane at the centerline of the simulation domain at t = 50 s. (b) The distribution of J^* at different porosity at t = 50 s. All simulation snapshots were taken on the X–Z plane at the centerline of the simulation domain at t = 50 s. (c) The total number of nuclei vs time. The total nuclei result from five sets of individual simulations are averaged. The error bar indicates the fluctuation range of the data. (d) The average nucleation rate vs time.

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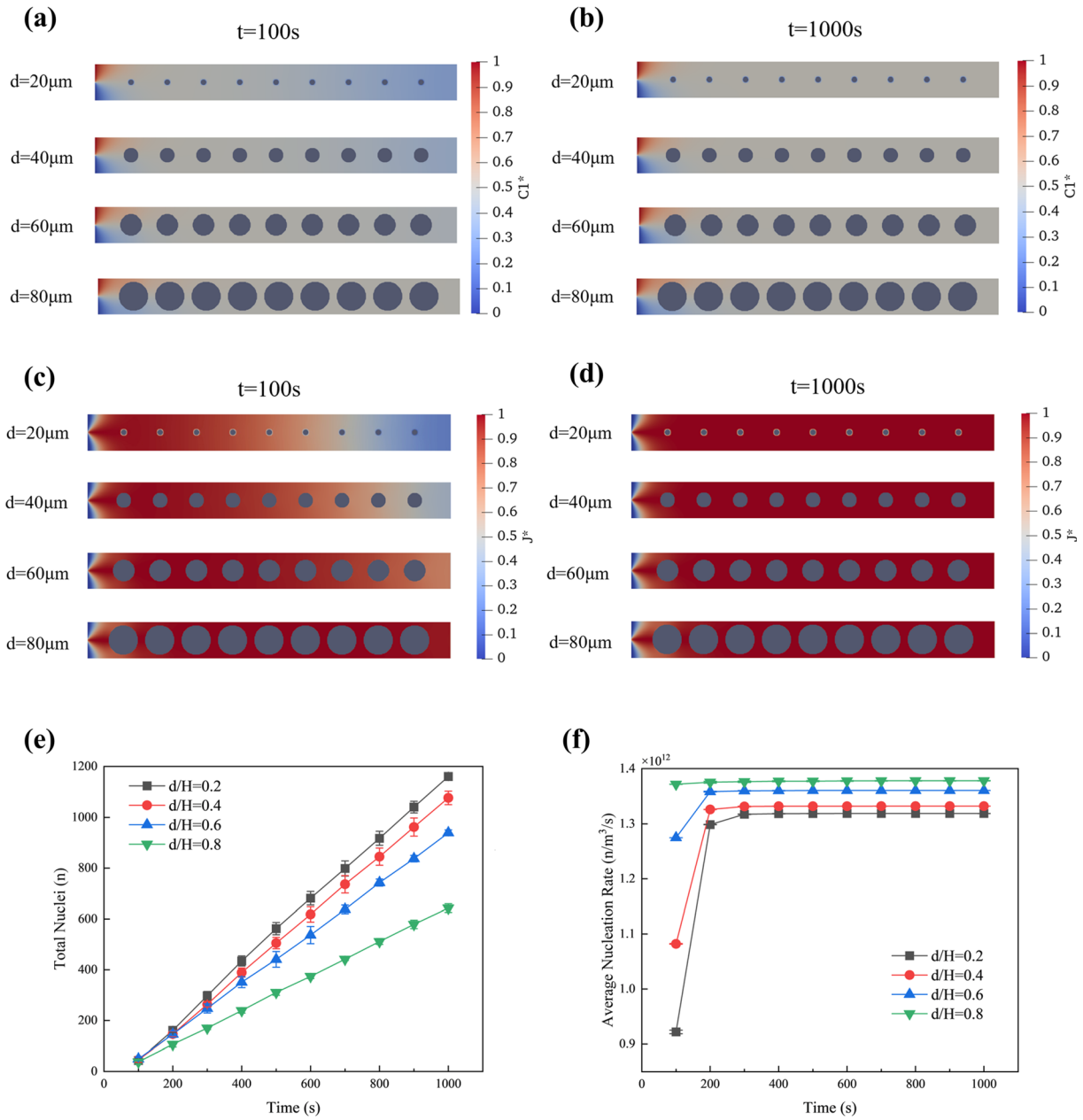


FIG. 8. (a) and (b) The distribution of C_1^* at different porosity (0.972, 0.887, 0.746, 0.548) at t = 100 s and t = 1000 s. All simulation snapshots were taken on the X–Z plane at the centerline of the simulation domain. (c) and (d) The distribution of J^* at different porosity (0.972, 0.887, 0.746, 0.548) at t = 100 s and t = 1000 s. (e) The total number of nuclei vs time. (f) The average nucleation rate vs time.

blue) forms with the smaller cylinders ($d = 20\ \mu$m). These results suggest that lower porosity cases enhance the mixing process and reach a steady concentration state more rapidly than those with higher porosity. Figures 8(c) and 8(d) show the distribution of J^* at different porosity at t = 100 s and t = 1000 s. We found that at t = 100 s, the nucleation rate distribution in the region under high porosity was uneven in the domain, while it was relatively uniform in the lower porosity cases. Such observation is consistent with the spatiotemporal evolution of ion concentration.

Figure 8(e) illustrates the temporal evolution of the total number of nuclei within the domain for different porosity. Even though the lower porosity cases have a higher average nucleation rate, their overall amount of nucleation occurrences gradually falls behind the higher

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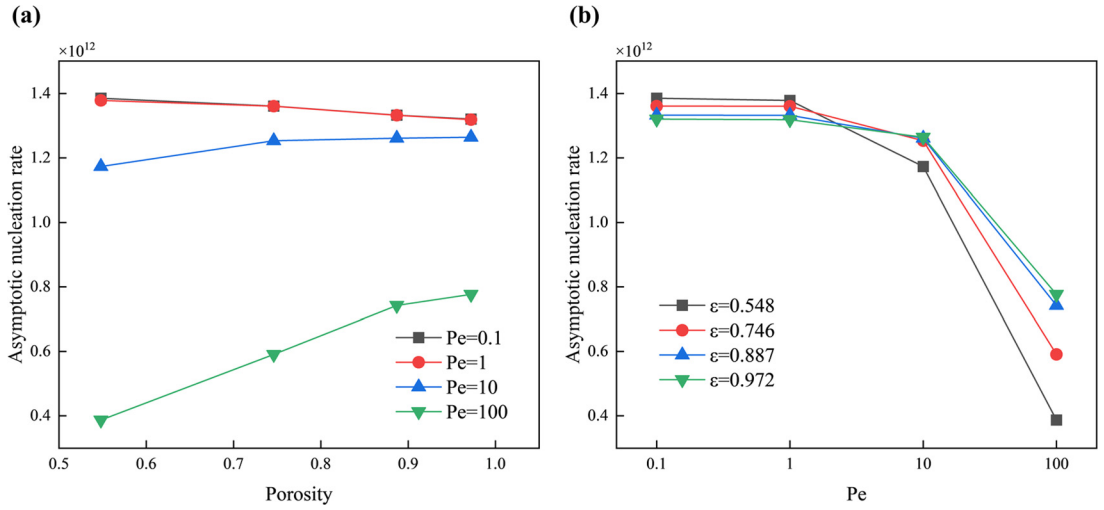


FIG. 9. (a) The evolution of the asymptotic nucleation rate in a porous matrix as a function of porosity at different *Pe* numbers (0.1, 1, 10, 100). The asymptotic nucleation rate represents the steady-state average nucleation rate throughout the simulation domain after the system stabilizes. (b) The evolution of the asymptotic nucleation rate in a porous matrix as a function of *Pe* at different porosities (0.548, 0.746, 0.887, 0.972). The ε represents the porosity of the system. The porosity ($\varepsilon = 0.548, 0.746, 0.887, 0.972$) corresponds to the cases (d/H = 80, 60, 40, and 20 μm).

porosity cases, which seems peculiar at first glance. This abnormal observation came from the fact that the overall volume available for homogeneous nucleation is much higher in the higher porosity cases, i.e., while the lower porosity cases provided a better mixing condition initially, their advantages faded as the transport in the higher porosity cases persisted. This trend could be better observed in Fig. 8(f), in which all cases eventually achieved similar levels of average nucleation rate with low *Pe* numbers.

The average nucleation rate approaches an equilibrium value over time [see Figs. 7(d) and 8(f)]. In other words, when the system reaches an approximate stability, its average nucleation rate also tends to stabilize. We define this average nucleation rate as the asymptotic nucleation rate, which reflects the nucleation efficiency of the system during its long-term evolution. Based on the original cases, we added relevant cases to explore the coupled influence of porosity and *Pe* number on the asymptotic nucleation rate. Figure 9(a) plots the evolution of the asymptotic nucleation rate in a porous matrix as a function of porosity at different *Pe* numbers (0.1, 1, 10, 100). It was found that at high *Pe* numbers ($Pe = 10$ and $Pe = 100$), the asymptotic nucleation rate increases significantly with the increase in porosity. This indicates that at high *Pe* numbers ($Pe = 10$ and $Pe = 100$), the increase in porosity can greatly enhance the nucleation efficiency. Conversely, at low *Pe* numbers ($Pe = 0.1$ and $Pe = 1$), the change in porosity has a weak effect on the asymptotic nucleation rate. This indicates that the effect of porosity on the asymptotic nucleation rate is highly dependent on the flow state. Figure 9(b) illustrates the evolution of the asymptotic nucleation rate in a porous matrix as a function of *Pe* at different porosity (0.548, 0.746, 0.887, 0.972). In the low *Pe* cases ($Pe = 0.1$ and $Pe = 1$), the range of variation of the asymptotic nucleation rate is relatively limited. This phenomenon once again confirms that when in the diffusion-dominated regime, the influence of porosity on the nucleation rate is relatively weak. However, in the high *Pe* case ($Pe = 100$), the asymptotic nucleation rate values show significant differences. At this point, the influence of porosity becomes particularly crucial. We also found that when *Pe* increased from 1 to 10, the asymptotic nucleation rate under different porosity began to change significantly, but the changes were not yet intense. This corresponds to the transitional stage of the physical mechanism from diffusion dominance to convection dominance. This change is also correspondingly reflected in Fig. 9(a). The alteration of this mechanism is mainly manifested as a significant increase in the sensitivity of the asymptotic nucleation rate to the change in porosity.

C. Effect of structure of porous matrix

To investigate the effect of the structure of the porous matrix on homogeneous nucleation, we changed the arrangement of cylindrical solids by adjusting the distance X to the centerline of the channel. Here, we fixed the diameter *d* of the cylindrical solids. By changing the X, we tuned the tortuosity of the porous medium to explore its effect on the homogeneous nucleation of minerals under different flow conditions. First, we set the fluid inlet velocity as $U = 1 \times 10^{-3}\,\mathrm{m\,s^{-1}}$, which corresponded to $Pe = 100$. Figure 10(a) shows the normalized concentration C_1^* at t = 50 s for different tortuosity. It is found that, when X = 0 μm, the mixing effect is relatively weak, and the overall concentration presents a clear vertical separation by the centerline. When cylindrical solids are arranged deviating from the central axes (X = 10 μm and X = 20 μm), which is equivalent to introducing asymmetry into the flow system, the disturbance of the concentration field is greatly enhanced. As the fluid bypasses these cylindrical solids, the solids will divert the fluid flow on both sides, which further strengthens the transport of species in the y-direction. Thus, this highly tortuous structure enhances the disturbance of the flow field and promotes the thorough mixing of solutes. Figure 10(b) displays the corresponding normalized nucleation rate J^* at t = 50 s. It is observed that, as X increases, the total area covered by the high nucleation rate increases,

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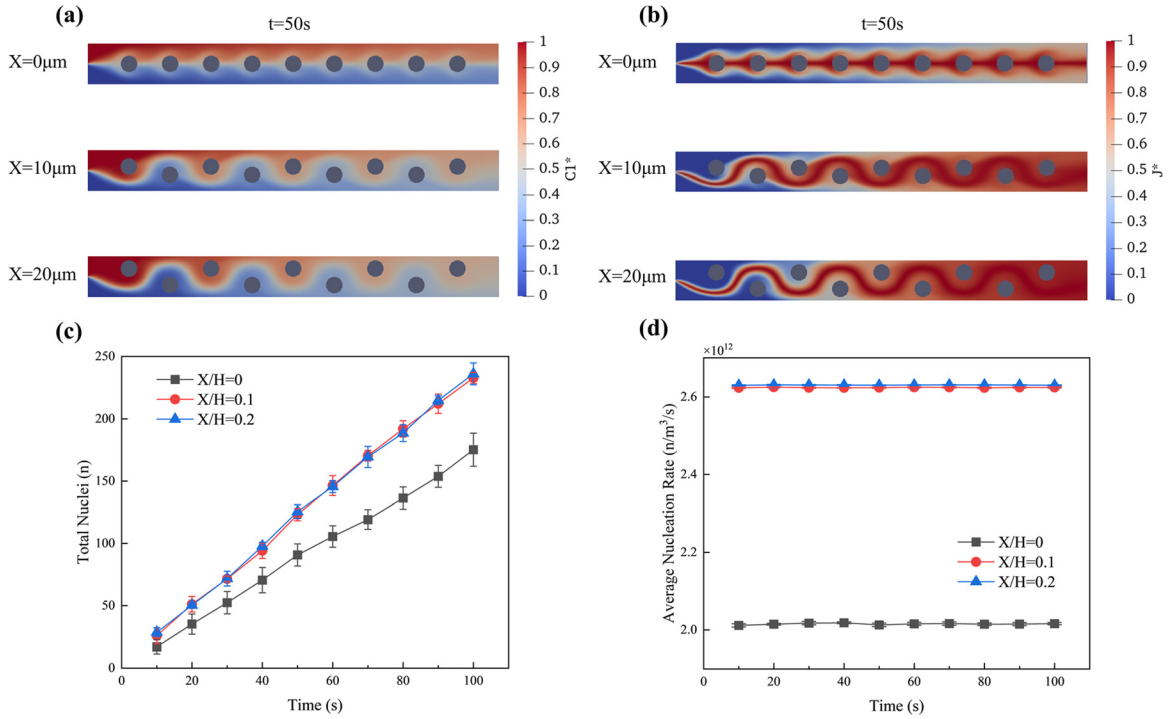


FIG. 10. (a) The distribution of C_1^* at different tortuosity at t = 50 s. Case 1 (X = 0 μm): Nine cylindrical solids (d = 40 μm) are uniformly arranged along the X–Z central axis of the simulation domain, corresponding to a tortuosity of $\tau_1 = 1.308$. Case 2 (X = 10 μm): Nine cylindrical solids deviate from the X–Z central axis of the simulation domain by 10 μm in sequence, corresponding to a tortuosity of $\tau_2 = 1.414$. Case 3 (X = 20 μm): Nine cylindrical solids deviate from the X–Z central axis of the simulation domain by 20 μm in sequence, corresponding to a tortuosity of $\tau_3 = 1.696$. All simulation snapshots were taken on the X–Z plane at the centerline of the simulation domain at t = 50 s. (b) The distribution of J^* at different tortuosity at t = 50 s. (c) The total number of nuclei vs time. (d) The average nucleation rate vs time.

and the distribution of J^* exhibits more complex spatial differentiation characteristics.

Figure 10(c) shows the increase in total nuclei as a function of time under different tortuosity. We found that, for X = 10 μm and X = 20 μm, the number of total nuclei in the simulation domain was significantly higher than in the case without deviation of position. This result is further supported by the plot of average nucleation rate [see Fig. 10(d)], which indicates that the structure significantly affects the homogeneous nucleation behaviors in the porous matrix. The above results demonstrate that high tortuosity enhances mixing and flow instability, thereby promoting homogeneous nucleation in a highly convective regime.

In contrast, when the advection is reduced ($Pe = 1$), the diffusion becomes the dominant transport mechanism, which causes the concentration field to be more uniformly distributed [see Figs. 11(a) and 11(b)] as well as the normalized nucleation rate [see Figs. 11(c) and 11(d)]. Such transport behavior is similar to what we observed for different *Pe* cases in microchannels (see Sec. IV A). Moreover, both the total amount of nuclei and average nucleation rate with different X are converged to the same level as shown in Figs. 11(e) and 11(f), which is expected as the concentration distribution does not vary significantly. In these cases, it is considered that the diffusion effect effectively eliminates the influence of the tortuous structure on the concentration gradient, since the diffusion characteristic time $\tau_{diff} = L^2/D$ (L is the channel length in the simulation) is much shorter than the nucleation observation window ($\tau_{diff} \ll 1000$ s). As a result, any local delay in solute transport caused by tortuous structures is rapidly compensated by diffusion, maintaining uniform supersaturation. In general, when the *Pe* number is relatively low, the nonuniformity of the concentration field is suppressed by the diffusion effect; therefore, the variation in the homogeneous nucleation behavior is insignificant.

We add relevant cases to explore the coupled influence of tortuosity and the Pe number on the asymptotic nucleation rate. The asymptotic nucleation rate is defined as the average nucleation rate in the entire simulation area after the system reaches stability, and it is a key indicator for measuring the overall efficiency of the homogeneous nucleation process. Figure 12(a) plots the evolution of the asymptotic nucleation rate in a porous matrix as a function of tortuosity at different *Pe* numbers ($Pe = 0.1, 1, 10, 100$). It was found that at low *Pe* numbers ($Pe = 0.1$ and $Pe = 1$), the asymptotic nucleation rate varies little with the tortuosity. This is mainly because in a diffusion-dominated system, the diffusion effect can quickly compensate for the transmission delay caused by the tortuous structure, thereby making the concentration field distribution tend to be uniform. At the convection-dominated system, the asymptotic nucleation rate increases significantly with the increase in tortuosity. This is because the highly tortuous structure promotes solute mixing and the formation of local supersaturated regions by disturbing the flow field, thereby enhancing the homogenization nucleation rate. We also found that when *Pe* increased from 1 to 10, the asymptotic nucleation rate under different

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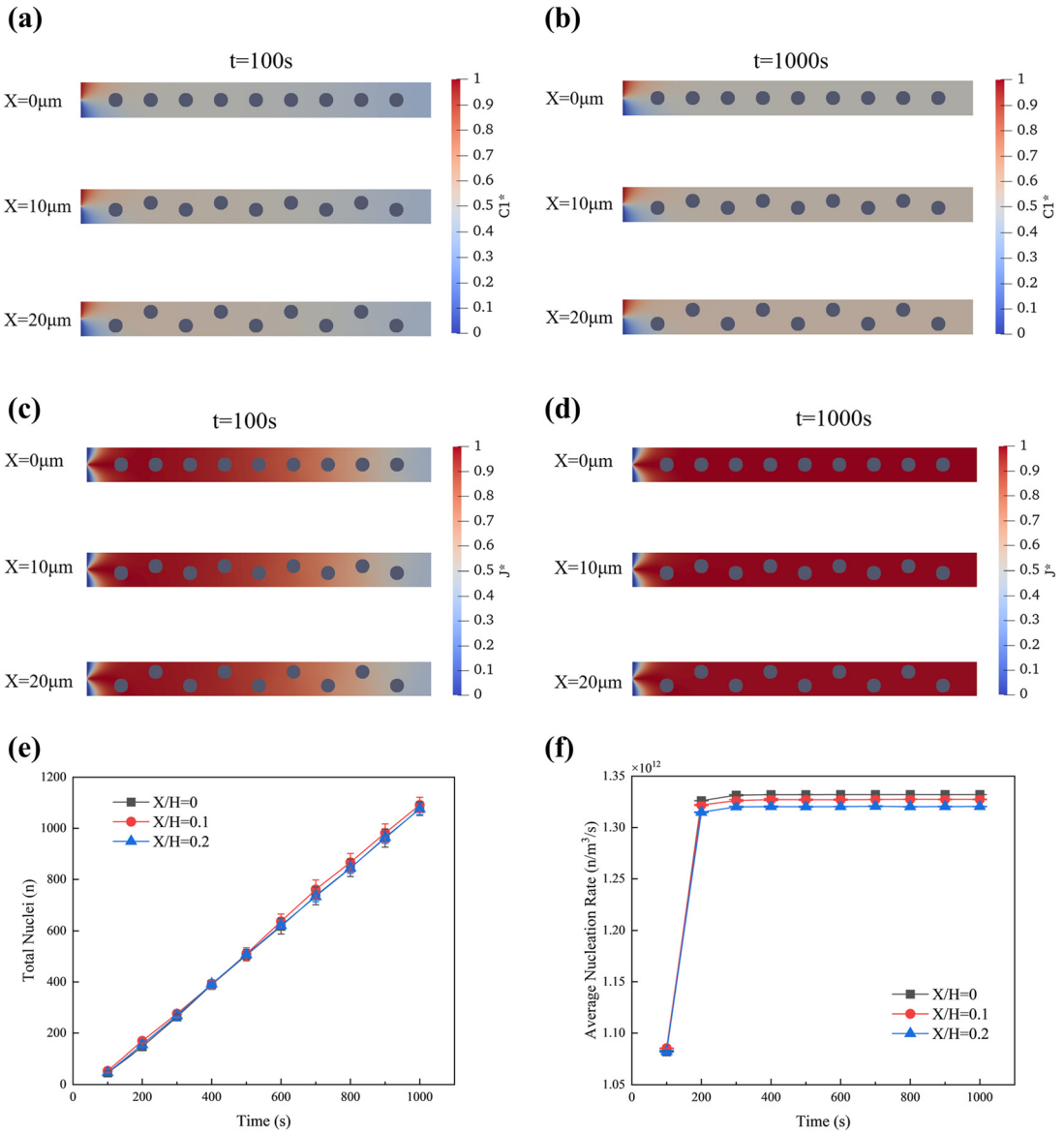


FIG. 11. (a) and (b) The distribution of C_1^* at different tortuosity (1.308, 1.414, 1.696) at t = 100 s and t = 1000 s. (c) and (d) The distribution of J^* at different tortuosity (1.308, 1.414, 1.696) at t = 100s and t = 1000 s. (e) The total number of nuclei vs time. (f) The average nucleation rate vs time. The error bar indicates the fluctuation range of the data.

porosity began to change significantly, but the changes were not yet intense. This corresponds to the transitional stage of the physical mechanism from diffusion dominance to convection dominance in the simulation. Figure 12(b) shows the evolution of the asymptotic nucleation rate in the porous matrix as a function of *Pe* at different tortuosity ($\tau = 1.272, 1.414, 1.555, 1.696$). We found that, at different tortuosity cases, the asymptotic nucleation rate first increases and then decreases with the increase in the *Pe* number. It indicates that the influence of tortuosity on homogeneous nucleation is related to the fluid flow state. It is only significant in the convective dominant regime (high *Pe*), promoting nucleation through enhanced mixing. The impact can be ignored under the diffusion-dominated system (low *Pe*).

V. CONCLUSION

In this study, a novel pore-scale model was developed for solving the complex reactive flow coupled with the stochastic homogeneous nucleation process—a process that was typically neglected in previous studies. Subsequently, we conducted rigorous verification of the homogeneous nucleation module and particle flow module by comparing simulation results with analytical solutions, thereby confirming the solver's reliability, accuracy, and stability. Then we constructed the

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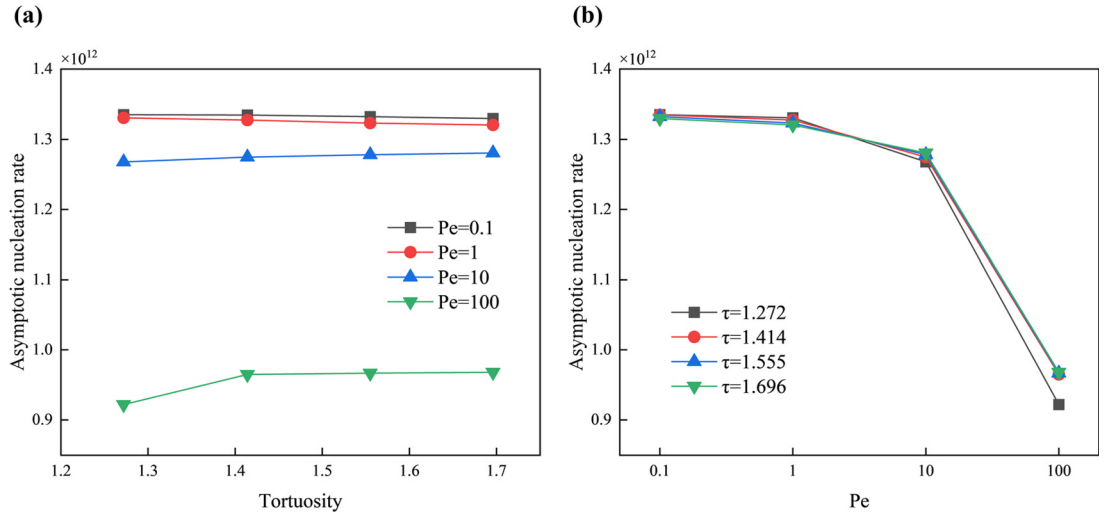


FIG. 12. (a) The evolution of the asymptotic nucleation rate in a porous matrix as a function of tortuosity at different Pe numbers (0.1, 1, 10, 100). The asymptotic nucleation rate represents the steady-state average nucleation rate throughout the simulation domain after the system stabilizes. (b) The evolution of the asymptotic nucleation rate in a porous matrix as a function of *Pe* at different tortuosity (1.272, 1.414, 1.555, 1.696). The τ represents the tortuosity of the system. The tortuosity ($\tau = 1.272, 1.414, 1.555, 1.696$) corresponds to the cases (X = 5, 10, 15, and 20 μm).

microchannel model and the porous medium model and systematically studied the influence of key control parameters such as *Pe* number, porosity, and tortuosity on the homogeneous nucleation.

The simulation results show that homogeneous nucleation behavior is strongly dependent on both the fluid flow conditions and the porous media structure. When the *Pe* number falls within the moderate range ($Pe = 1$–10), where advection and diffusion are in balance, the average nucleation rate in the microchannel system is the highest, resulting in optimal nucleation efficiency. The effects of porosity and tortuosity of porous media on mineral homogeneous nucleation are flow-regime dependent. High porosity consistently provides more volume for nucleation, thereby increasing the total nucleation amount, particularly in the advection-dominated regime (high *Pe*). In addition, the effect of tortuosity is prominent only in the advection-dominated regime (high *Pe*), where highly tortuous structures enhance mixing by disturbing the flow field and promote nucleation; its influence is negligible in the diffusion-dominated regime (low *Pe*).

Understanding and predicting the evolution of Earth's systems resulting from mineral nucleation and precipitation are important. In actual geological systems, there are numerous types of minerals, and the temperature and pressure conditions are complex and variable. The governing equations of this model (e.g., fluid flow in porous media and particle dynamics) have wide applicability. Therefore, the key to expanding its application scope lies in obtaining and inputting the precise thermodynamic and kinetic parameters of the corresponding minerals (e.g., calcite and celestine). Our study provides an important quantitative reference index (*Pe*) for optimizing the injection strategy and achieving efficient and safe CO_2 sequestration and clarifies the specific optimization direction, that is, to actively regulate the local flow state of the reservoir to keep it in the optimal range of diffusion and convection balance ($Pe = 1$–10). Within this ideal range, convection and diffusion can achieve the best synergistic effect, thereby realizing homogeneous nucleation behavior within the maximum range and forming a large number of carbonate crystal nuclei, laying a solid foundation for long-term and stable CO_2 sequestration.

Note we did not delve into the complex dynamic behavior of crystal particles (e.g., crystal growth, clogging effect) as they are not the focus of the current study. However, in future studies, it is worth considering the crystal growth process and the interactions between crystal particles as well as the evolving porous matrix, while incorporating more realistic pore geometries for in-depth investigation. In this study, we mainly focused on the homogeneous nucleation of minerals and did not involve the heterogeneous nucleation process. Future research will also focus on exploring the competitive relationship between homogeneous nucleation and heterogeneous nucleation of minerals. The outcomes of this study are anticipated to provide insights into the mechanisms of mineral homogeneous nucleation and precipitation in porous media.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Yue Zou: Conceptualization (lead); Data curation (equal); Formal analysis (equal); Investigation (equal); Project administration (lead); Writing – original draft (equal); Writing – review & editing (equal).

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DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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